

Symmetric Encryption

Classic Ciphers

Computer Systems Security

2025/26

LEI - UM

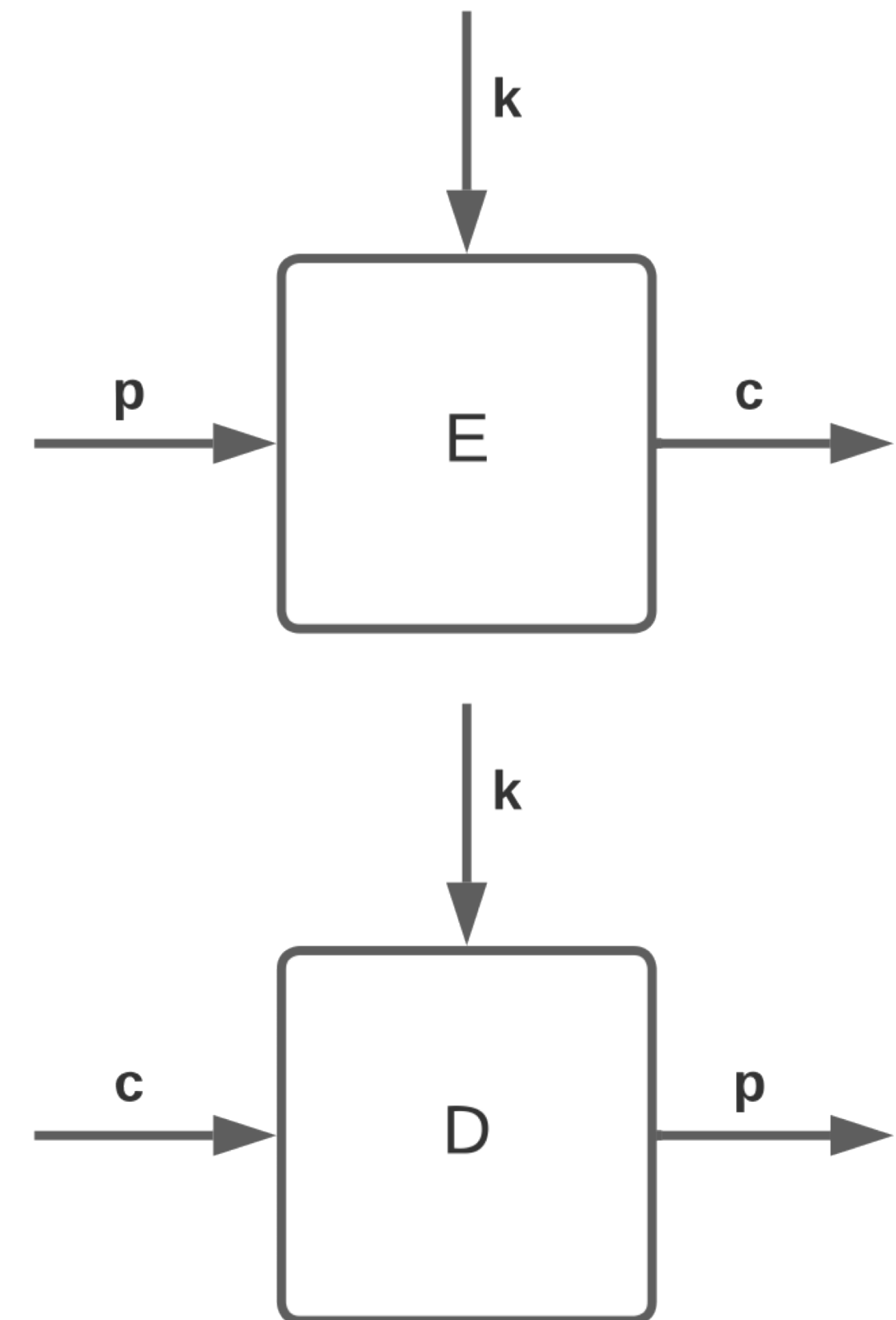
Hugo Pacheco hpacheco@di.uminho.pt

Context

- People usually think of encryption when they hear about cryptography
 - Not unreasonable, but only one of many cryptographic techniques
- Encryption guarantees confidentiality
 - Many other security properties are required in real-world systems
 - Authenticity, non-repudiation, unpredictability, anonymity, ...
- There are many types of encryption
 - Symmetric, asymmetric, authenticated, homomorphic, ...

Encryption

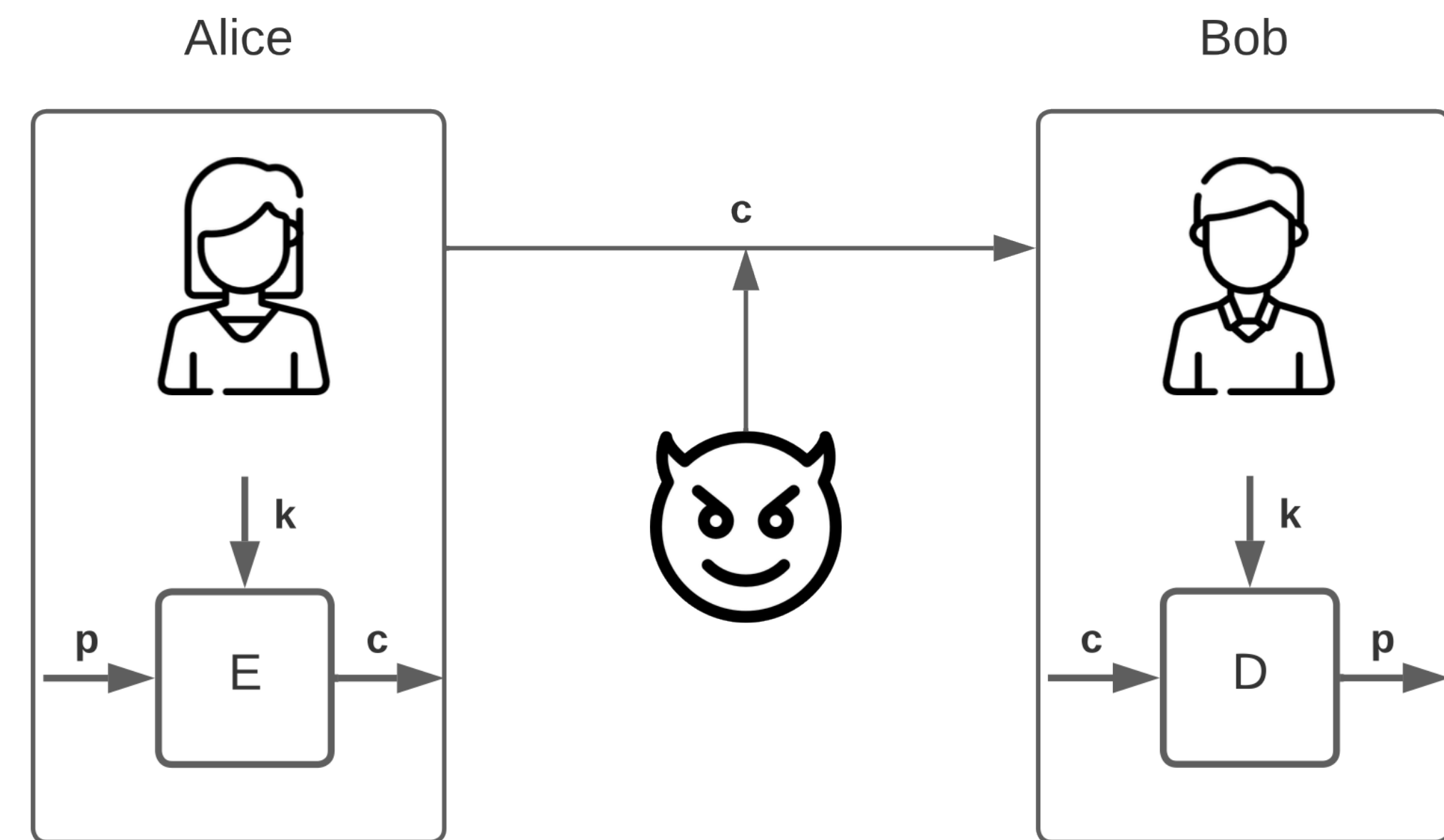
- Encryption transforms plaintexts into ciphertexts using a key
 - Encryption is (usually) randomized
 - $c \leftarrow_{\$} E(k, p)$
 - Decryption is (necessarily) deterministic
 - $p \leftarrow D(k, c)$
- We begin with **symmetric encryption**
 - Same key on both ends



Symmetric Encryption

Security

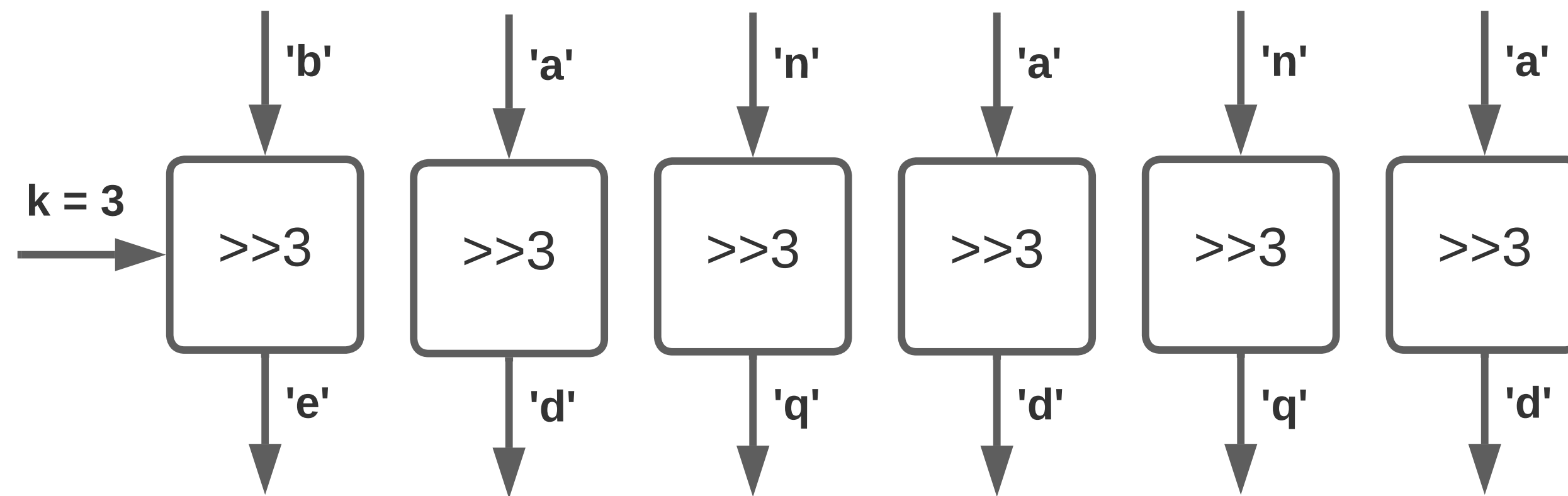
- Alice wants to send a message to Bob
 - They agree on a key beforehand
- The attacker controls the communication channel
- What does it mean for the encryption to be secure?
 1. The exact message is not revealed?
 2. No character of the message is revealed?
 3. No pattern of the message is revealed?
 4. No information of the message is revealed?



Caesar Cipher

- One of the earliest known ciphers, known to have been used by Julius Caesar during the Gallic campaign

- Encryption is a shift of the letters of the alphabet



- Example: Encrypting the message CartagoEstaNoPapo with key $K=6$ results in IGXZGMUKYZGTUUVGVU
- How can we decrypt? Perform the shift in the opposite direction

Caesar Cipher

Security

- Key is fixed for encryption / decryption
- Why is this cipher **insecure**?
 - The total number of possible keys is 26 (one of which is weak)
 - **Very small key space!**
- E.g. want to attack the ciphertext: FXLNTQCLOPNPDLC
- We can try a simple strategy:
 1. decrypt with every possible key
 2. spot the one(s) that “make sense”

CYPHERTEXT:		FXLNTQCLOPNPDLC
+1	:	GYMOURDMPQOQEMD
+2	:	HZNPVSENQRPREFNE
...	:	
[MSG] +15 (K=26-15=11)	:	UMACIFRADECESAR
...	:	
+24	:	DVJLROAJMNLNBJA
+25	:	EWKMSPBKNOMOCKB

Caesar Cipher

Brute force attack

- The previous strategy is known as a **Brute-force Attack**
the adversary searches the entire key space in the hope of finding the right key
- To be effective, it assumes that:
 - there is enough redundancy in the original message
 - or that a plaintext/ciphertext pair is known
- An attack that can always be applied to a cryptosystem
 - but whose feasibility depends on the size of the key space
- **Key sizes are a necessary criterion for the security of ciphers**

Ciphers

On key sizes

- If we consider keys to be arbitrary bitstrings, the size of the key space (2^n) increases exponentially on the key size (n)

- Example:

Key Size	Time (1 μ sec/test)	Time (1 μ sec/10 ⁶ tests)
32 bit	35.8 min.	2.15 msec.
40 bit	6.4 days	550 msec
56 bit	1140 years	10 hours
64 bit	500000 years	107 days
128 bit	5 * 10 ²⁴ years	5 * 10 ¹⁸ years

- Baseline: 2^{128} provides a reasonable security level!

Monoalphabetic Substitution Cipher

Increasing the key size

- Instead of the fixed shift used in the Caesar Cipher, we can choose different shifts for different letters

- E.g.

A	B	C		W	Y	Z
R	X	K		B	I	F

- We are defining a permutation over the alphabet
- How large is the key space?
 - We have $26! \approx 2^{88}$ permutations
 - A big number, not possible to brute force without massive investment
 - **Should we trust the security of this cipher?**

Monoalphabetic Substitution Cipher

Frequency analysis attack

- Which of these is most common in Portuguese?

L	A	S	Z
2.78%	14.63%	6.81%	0.47%

- How can we turn this into a strategy?
 1. Gather many ciphertexts and count the frequency of letters
 2. Match that frequency with the frequency of plaintext alphabet
- With good odds, the most common letter in the ciphertexts will match the most common letter in the plaintext alphabet

Monoalphabetic Substitution Cipher

Frequency analysis attack

- Consider the following ciphertext

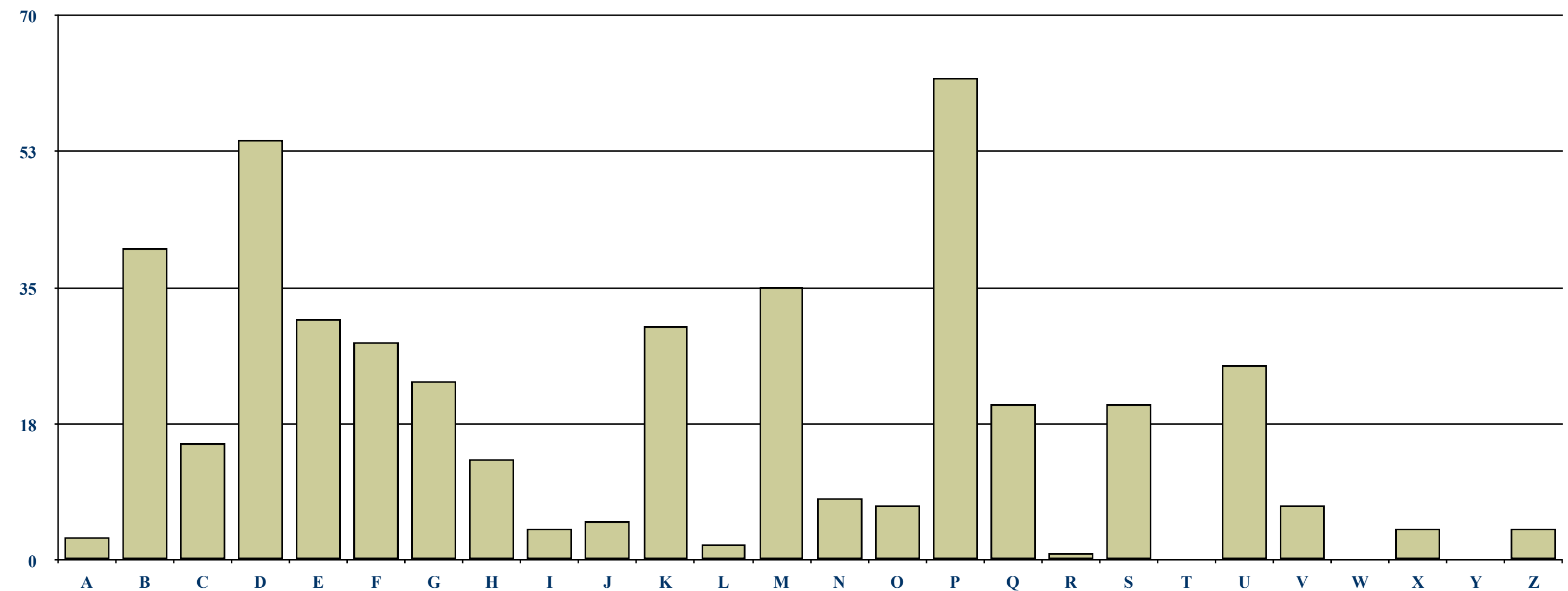
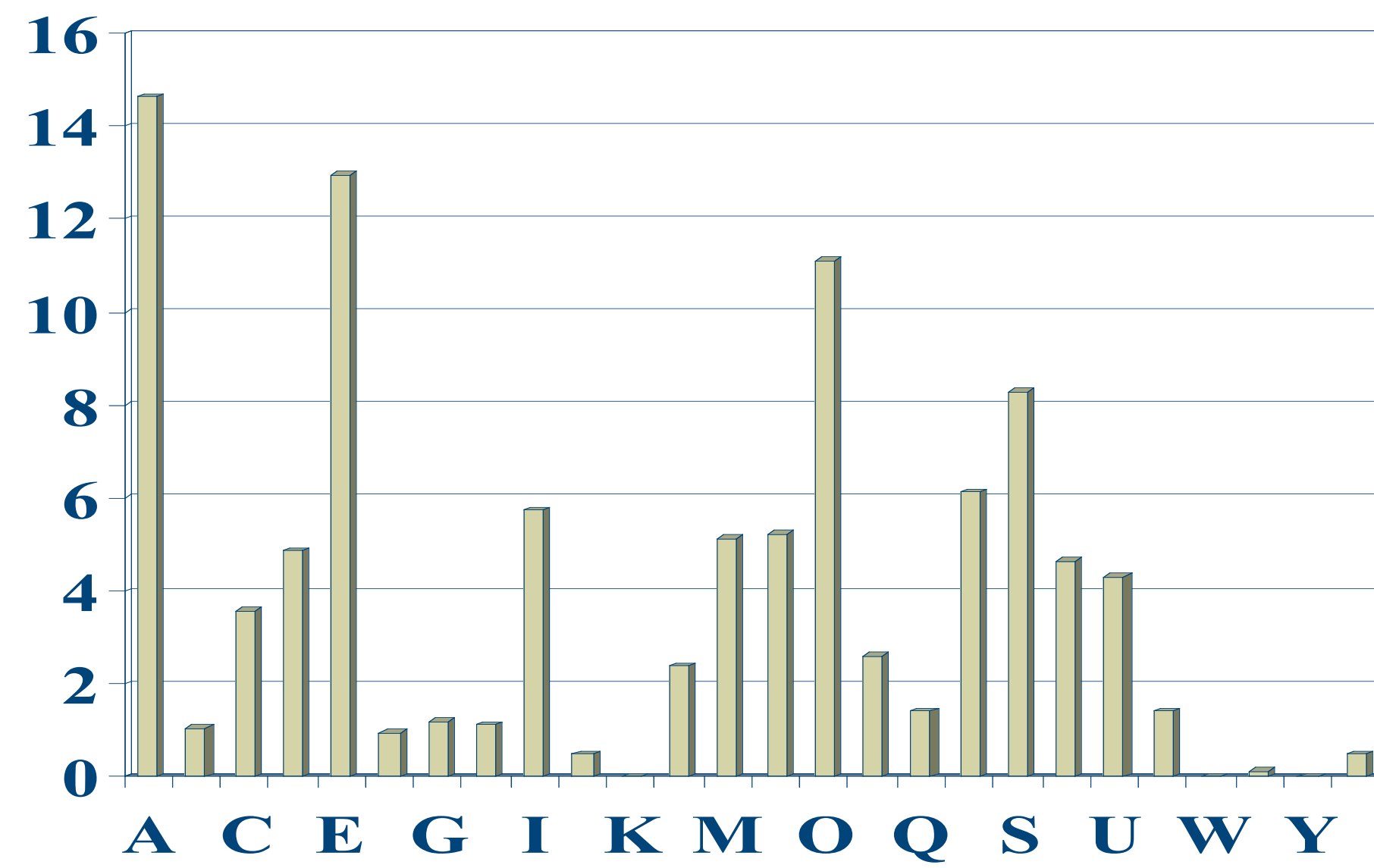
FPGFBNBVPKFBDMSBEMDMGUCDKDGUGDMUSPMMDBEFLEFEQDCPPGIDEXDCBKPMDFQBUGPSUGHKEGPF
QBMPXPKSESEBSURBHKHBMEQBFUFSDSBGHKPPFCECPHDKQPDHDKQPFDBADVEDFDCDDCEZPKLDZEDGMPPM
NDKPMGDVPEMPDNUPFQDVDMPCPZGEFUQBMCPUGMEOPFSEBHPFBMBFBDUNPCDPXSEQDSDBCBBUQBFBCPMU
KNEKDGUGDMHPKMBFDNPFMCPBKENPGBIMSUKDSBGJUPGPFQKPQEVPSBFSEOEDIUOBMPGKPMQDUKDFQPMPX
SPFQKESBMPMMPMQKDZEDGUGDHPKNUFQDQPKKEVPOBJUPPJUPVDEMUSPCPKKPMHBFCEOAPMJUPFDBMDIED
PPOPMBOADKDGHDKDBHKDQBSBGEFJUEPQDSDB

- Known to be a Portuguese text
- How should we proceed?

Monoalphabetic Substitution Cipher

Frequency analysis attack

- Letter frequency of Portuguese
- Histogram of letters in the ciphertext



Monoalphabetic Substitution Cipher

Frequency analysis attack

- Suggests matching “A”, “E”, “O” with “P”, “B”, “D”
- Pairs (bigrams) or triples (trigrams) of letters can also be considered, such as “os”; “es”; “que”; “nao”; ...
- Several occurrences of the pairs “PM”, “PF”, “MP”, and “JUP” suggest the following partial decryption: {...; **F:M**; ...}

ME-MO-O-E-MOAS-O-SAS-U-A-A-U-ASU-ESSAO-M--M--A-EE--A--A-O-ESA--ESEM-OU-E-U--
---EM-OSE-E-----O-U-O--O-OS--OMUM-A-O---EEM---E-A--EA-A--EMAO-A--AMA-AA---E-
-A--A-SEES-A-ESA-A-E-SEA-UEM-A-ASE-E---MU-OS-EU-S--EM--O-EMOSOMOAU-E-AE-----
A-AO-OOU-OMO-ESU----A-U-AS-E-SOMA-EMS-EO---E-O-S-U-A-O-QUE-EM--E---E-OM-----
A-U-OSE--ES-AU-AM-ESE--EM----OSESES--A--A-U-A-E--UM-A-E----E-OQUEEQUE-A-SU-
E-E--ES-OM----ESQUEMAOSA--AEE-ESO--A-A--A-AO--A-O-O--MQU-E-A-AO

Monoalphabetic Substitution Cipher

Frequency analysis attack

- We need to backtrack on some of our hypotheses...
- Reassigning the decryption of F into N
 - We get something a lot more promising

NE-NO-O-ERNOAS-O-SAS-U-ARA-U-ASU-ESSAO-N--N-TA-EE--A--A-ORESA-RESENTOU-E-U--R--
ENTOSE-ER----O-U-O-RO-OS-TONUN-A-O--REEN---E-ARTEA-ARTENAO-A--ANA-AA---ER-A--A-
SEES-ARESA-A-E-SEA-UENTA-ASE-E---NUTOS-EU-S--EN--O-ENOSONOAU-E-AE---TA-AO-
OOUTONO-ESUR--RA-U-AS-ERSONA-ENS-EOR--E-O-S-URA-O-QUE-ENTRET--E-ON---A-U-OSE-
RESTAURANTESE--ENTR--OSESSESTRA--A-U-A-ER-UNTATERR--E-OQUEEQUE-A-SU-E-ERRES-ON---
-ESQUENAOSA--AEE-ESO--ARA--ARAO-RATO-O--NQU-ETA-AO

Monoalphabetic Substitution Cipher

Frequency analysis attack

- Which rapidly leads to the whole plaintext

NEMNOGOVERNOASCOISASMUDARAMUMASUCESSAOINFINITADEEMBAIXADORESAPRESENTOUMECUMPRIMEN
TOSEXERCICIOCUJOPROPOSITONUNCACOMPREENDEPARTEAPARTENAOHAVIANADAADIZERFAZIAMSEES
GARESAMAVEISEAGUENTAVASEDEZMINUTOSDEUMSILENCIOPENOSONOAUGEDAEXCITACAODOOUTONODESU
RGIRAMUMASPERSONAGENSDEORIGEMOBSCURACOMQUEMENTRETIVECONCILIABULOSEMRESTAURANTESEX
CENTRICOSESSESTRAZIAMUMAPERGUNTATERRIVELOQUEEQUEVAISUCEDERRESPONDILHESQUENAOSABIA
EELESOLHARAMPARAOPRATOCOMINQUIETACAO

Vigenère Cipher

Polyalphabetic substitution cipher



- Attributed to *Blaise Vigenère* (16th century). Known as “*le chiffre indéchiffrable*”
- It interleaves multiple Caesar ciphers
 - Key is a password — each letter corresponds to a single Caesar cipher key (A=0; B=1; ...)
 - Encryption applies the Caesar cipher with each character of the key in sequence, starting over when no more key characters are available
- Same letter is not always encrypted in the same way (add some hurdles to the frequency analysis)
 - **But if the plaintext is much larger than the key, patterns in the plaintext shall be reflected in the ciphertext**
- Broken in the 19th century by *Charles Babbage* and *Friedrich Kasiski*
 - The employed cryptanalysis techniques already exhibited some degree of sophistication

Rotor Machines

- Hebern machine
 - Key is the disk, encoding a substitution table
 - On key press, the output is encrypted and the disk rotates



- The Enigma
 - Key is the initial setting of rotors by multiple rotors (3-5)
 - Rotors rotate with different frequencies



Transposition Cipher

- Encryption permutes character positions
- E.g.
 - Consider a permutation [2; 1; 3] (key)
 - Encrypt “AindaOutraCifra”
 1. Write it on a matrix
 2. Read the columns (with header [1; 2; 3])
 - Resulting ciphertext: “IATCRADUAFNORIA”
- **Frequency analysis is now useless!**

2	1	3
A	I	N
D	A	O
U	T	R
A	C	I
F	R	A

Cipher Composition

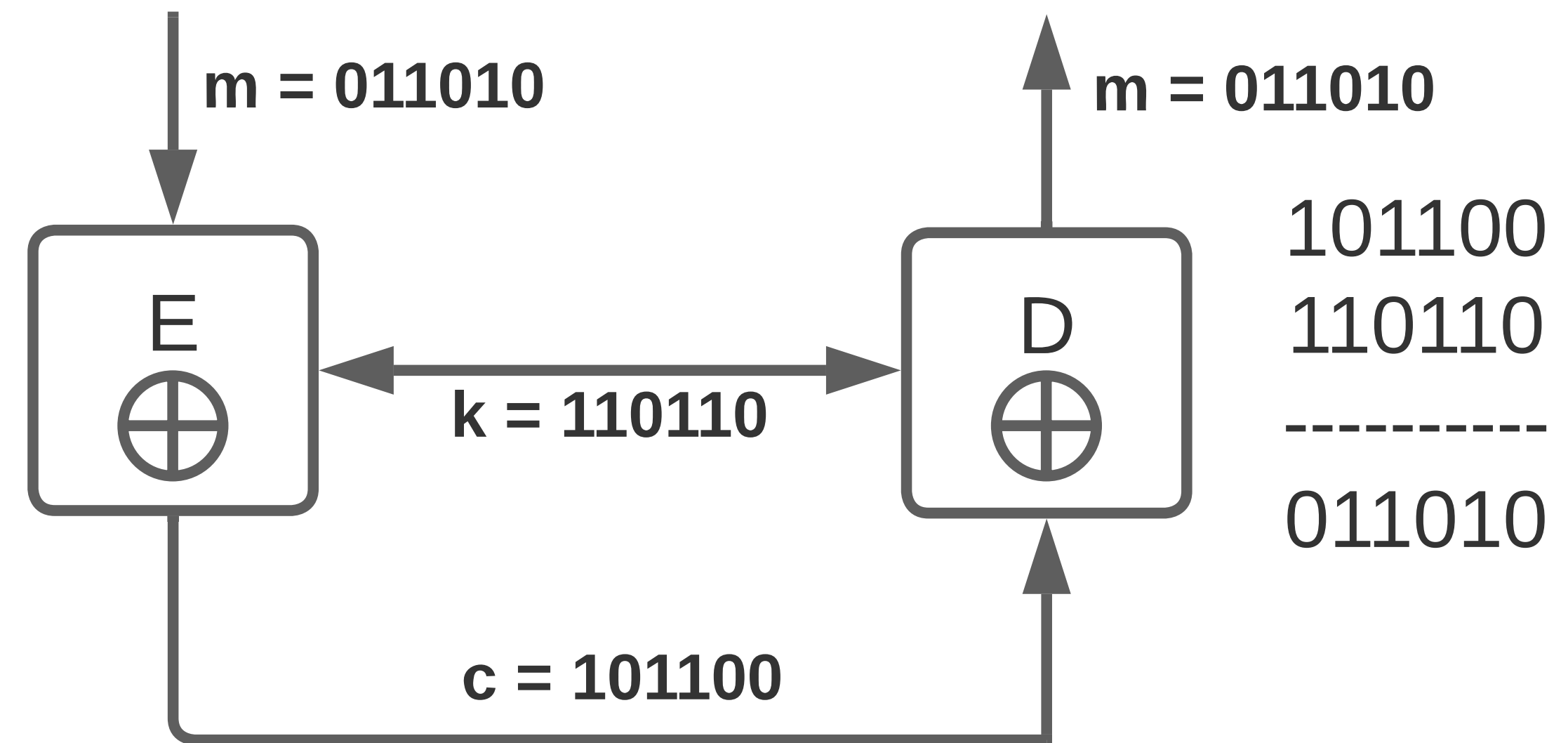
- We have seen different (simple) ciphers
- Can we combine multiple ciphers to make them more secure?
 - **It depends!**
 - It may happen that it doesn't add any value (e.g., combining two substitution ciphers)
 - However, some combination patterns can be advantageous (e.g., by interleaving substitutions with permutations — a.k.a. Substitution-Permutation networks or SP-networks)
 - Actually used in modern ciphers

One-Time Pad

Vernam Cipher - 1917

011010
110110

101100



- Generalises Vigenère cipher with:
 1. key size is (at least) the plaintext size
 2. key is fully random
 3. key is only used in a single encryption
- Normally described as operating on a binary alphabet (0s and 1s)
- Encryption/decryption is the *exclusive-or* (XOR) with the corresponding key bit

$$C_i = T_i \oplus K_i$$

$$M_i = C_i \oplus K_i$$



One-Time Pad

- Shown to be secure (**information-theoretically secure**) by *Claude Shannon* — “*knowing the ciphertext does not reduce the uncertainty about the plaintext*”
- Why is this not used to encrypt everything?
 - Keys must have the same size as the messages
 - E.g., to send a 2GB file I must use a 2GB key
 - How can we pre-share and store such huge keys?
 - **Verdict: Key generation and distribution make this unfeasible in realistic scenarios**
 - When seen strictly as a cipher. Used in many constructions as a building block

Takeaways

- In cryptanalysis, all available information is used, including:
 - Partial information about the transmitted message
 - Previous cipher's use (e.g. messages encrypted with the same key)
 - Possible weaknesses in the cipher's use (e.g. deficiencies in the choice of keys, etc)
- Although unconditionally secure techniques exist, they often have strict requirements that make them impractical
- Most cryptographic techniques used today are based on **computational security**, which limits the adversary's computational capabilities
- The size of a cipher's key should be determined based on the desired level of security (e.g. 2^{112}), while taking into account the amount of key size that is “consumed” by known cryptanalysis techniques

On Randomness

- Which of these numbers is more random?
 - 00000000
 - 10101010
 - 00100100
 - 10011101

On Randomness

Statistical Tests

- Look for bit/byte distributions and patterns
- *Golomb's* Randomness Postulates
 1. The difference in the number of 1s and 0s must tend towards zero
 2. The expected number of sub-sequences of repeated symbols (runs) with length l is given by $r(l)=r/2^l$
 3. The auto-correlation must be a constant value for any deviation other than 0 ($\text{mod } p$)
- **Completely insecure for (most) cryptographic purposes!**
 - Is the number π random?

On Randomness

Distributions

- **Randomness is not a property of a bit string, but of the distribution!**
- Randomized processes are described using randomness distributions
 - A process U samples from the uniform distribution over a finite field S if

$$\forall s^* \in S, \Pr[s = s^* : s \leftarrow U] = \frac{1}{|S|}$$

- If we roll a fair dice, what is the probability of getting 1?
- If we sample a byte, what is the probability of getting 00000000 or 10011101?

On Randomness

Indistinguishability

- In cryptography, we are typically concerned about random number generators
 - Assume that we generate a sequence of random bits iteratively
 - Is the next bit **unpredictable** to an adversary, even if he has seen all previous bits?

- Perfect (information-theoretic) Security:

A sequence must be indistinguishable from a (truly random) uniform distribution

- Cryptographic (computational) Security:

A sequence must be indistinguishable from a uniform distribution for any PPT adversary

- Often generated from a truly random seed by a deterministic pseudo-random number generator (PRNG)

On Randomness

Entropy

- Entropy is a measure of **uncertainty**, measured in bits of randomness
 - Considers a choice from a set of possible outcomes uniformly at random
 - Calculated as $\log_2(\text{\#possibilities})$
- The more bits of randomness we have, the better off we are from a security standpoint
 - E.g., a coin has $\log_2(2) = 1$
 - E.g., a die has $\log_2(6) \approx 2.58$
 - E.g., a 5-digit password has $\log_2(10^5) \approx 16.61$